

Applied Machine Learning Lab

B.Tech Semester 4

Experiment 02

Data Preprocessing and Exploratory Data Analysis (EDA)

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AIM:

To perform Exploratory Data Analysis (EDA), clean and preprocess data, handle missing values, scale features, perform feature engineering, and apply dimensionality reduction techniques.

THEORY:

EDA helps understand data patterns. Missing values are handled using imputation. Outliers are detected using IQR. Feature scaling standardizes data. Feature engineering creates useful features. PCA reduces dimensionality.

Commands/Code Used:

```
from sklearn.datasets import load_iris

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import PCA

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)

df["target"] = iris.target


# Correlation heatmap

plt.figure(figsize=(8,6))

sns.heatmap(df.corr(), annot=True, cmap="coolwarm")

plt.savefig("eda_correlation.png")


# Distribution plots

df.hist(figsize=(10,8))

plt.savefig("eda_distributions.png")


# PCA

X = df.iloc[:, :-1]

y = df["target"]
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X_scaled)
```

```
plt.figure(figsize=(8,6))
```

```
plt.scatter(X_pca[:,0], X_pca[:,1], c=y, cmap="viridis")
```

```
plt.colorbar()
```

```
plt.title("PCA: Dimensionality Reduction")
```

```
plt.savefig("pca_reduction.png")
```

RESULTS:
Data distributions, correlations and PCA visualization were successfully generated.

CONCLUSION:

Data preprocessing and EDA were successfully performed and are essential for ML models.